STA5092Z 2026 Assignment 1

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I will doing some EDA on very messy data, yay!

Part 1- Data Wrangling

Load Data

2) Load data and transform into tibble objects

Rename columns in nest_all to match those to fledg_survival. In this case, columns 'GRP' and 'Group' mean the same thing in both dataframes, so 'GRP' is renamed to 'Group' in the nest_all data frame.

Data Exploration

3)

DataFrame Column Data Types

Key Findings before merging:

fledg_survival dataframe

- Total Number of Rows=427 rows
- Total number of unique Bird IDs=425
- Data types in columns are mostly split between character and int/numeric data types
- **Date** column in fledg_survival is stored as a character- will need to be converted. The data format is of the format dd-mm-yy using lubridate package
- **SurvMassAtInd** column contains null values

- **Group** which denotes which nest group a bird belongs to- however this is named differently in the nest_all dataframe

nest_all dataframe

- Total number of rows=596
- Total number of unique Bird IDs=586
- In addition to numeric and int/numeric data types, columns also include logical/boolean data types
- **Date** is also a character data type, similar to the format in the **fledg_survival** dataframe

The key differences between the datasets is summarized in the table below:

attribute	nest_all_data	fledg_survival_data
number of rows	596	427
number of columns	81	39
number of unique Bird IDs	586	199
number of unique Nest IDs	257	257

Pre Merging Data Preparation Steps:

1. Rename columns in nest_all to match those to fledg_survival. In this case, columns 'GRP' and 'Group' mean the same thing in both dataframes, so 'Group' is renamed to 'GRP' in the nest_all data frame.
2. Find common column names in both the **nest_all** and **fledg_survival** data sets to find columns to join on. Particulary, for the **Date** column, this will be convert date columns in both data frames into datetime columns using the lubridate ymd function
3. Check and ensure that the datatypes of these common columns are the same in both data sets- using the janitor package- and change any datatype mismatches to the same data type
4. Both Both GrpSizeAD and GrpSizeTotal columns are stored as characters in the nest_all data frame, as they both contain #NA values
5. These will replaced with na values instead, once they are coerced into integer values

Find Common columns and check their datatypes- highlighting the mismatches

```
 [1] "Date"              "BirdID"            "GRP"
 [4] "Season"            "AgeMeasD"          "Sex"
 [7] "NestID"            "GrpSizeAD"         "GrpSizeTotal"
[10] "BroodSize"         "NoMales"           "BroodSexRatio"
[13] "Mass"              "SurvInd"           "SurvMassAtInd"
```

```
[16] "Surv1Yr"            "Surv1Mass"          "MeanTmaxInc"
[19] "MeanTmaxBrood"      "MeanTmaxInd"        "MeanTminInc"
[22] "MeanTminBrood"      "MeanTmin90"         "MeanTvarInc"
[25] "MeanTvarBrood"      "MeanTvar90"         "HWaveInc"
[28] "NoHotDaysInc"       "PropHotDaysInc"     "HWaveBr"
[31] "NoHotDaysBr"        "PropHotDaysBr"      "NoHotDays90"
[34] "PropHotDays90"      "RainTwoMonthsPrior" "Rain90"
[37] "Drought"
```

Table 2: Data type mismatch between common columns in nest_all and fledge survival datasets

column_name	fledg_survival_common	nest_all_common
GrpSizeAD	integer	character
GrpSizeTotal	integer	character

Columns, **GrpSizeAD** and **GrpSizeTotal** are different data types in both data sets. In particular, both GrpSizeAD and GrpSizeTotal in the nest_all dataset contain string "#NA" values- so these are being read in as chartersColumns GrpSizeAD and GrpSizeTotal have different data types in both datasets. Specifically, both GrpSizeAD and GrpSizeTotal in the nest_all dataset contain the string "#NA" values, which means they are being read as characters.Columns GrpSizeAD and GrpSizeTotal have different data types in both datasets. Specifically, both GrpSizeAD and GrpSizeTotal in the nest_all dataset contain the string "#NA" values, which means they are being read as characters.

```
 [1] "6"    "8"    "7"    "3"    "5"    "4"    "9"    "10"   "11"   "#N/A"
[11] "12"   "2"

[1] "6"    "7"    "3"    "5"    "4"    "2"    "8"    "9"    "#N/A"
```

To correct for this, these columns are mutated as integer columns in the nest_all data set. Doing this will preserve the numerical values in the columns as integers, whilst the character values will be stored as NA which R will recognize as null values.

Merge Data Sets

nest_all contains more rows and unique NestIDs than the fledge_survival data set. The latter only contains data of birds that survived to fledge, whilst the former contains information on all birds- regardless of their survival status; fledge_survival is a subset of nest_all. As such, in order to preserve information from both data sets, the data sets will be merged using a left join- which preserves all information from nest_all whilst getting additional fledging information of birds that fleged from fledg_survival.

Part 2- Data Summaries and Visualizations

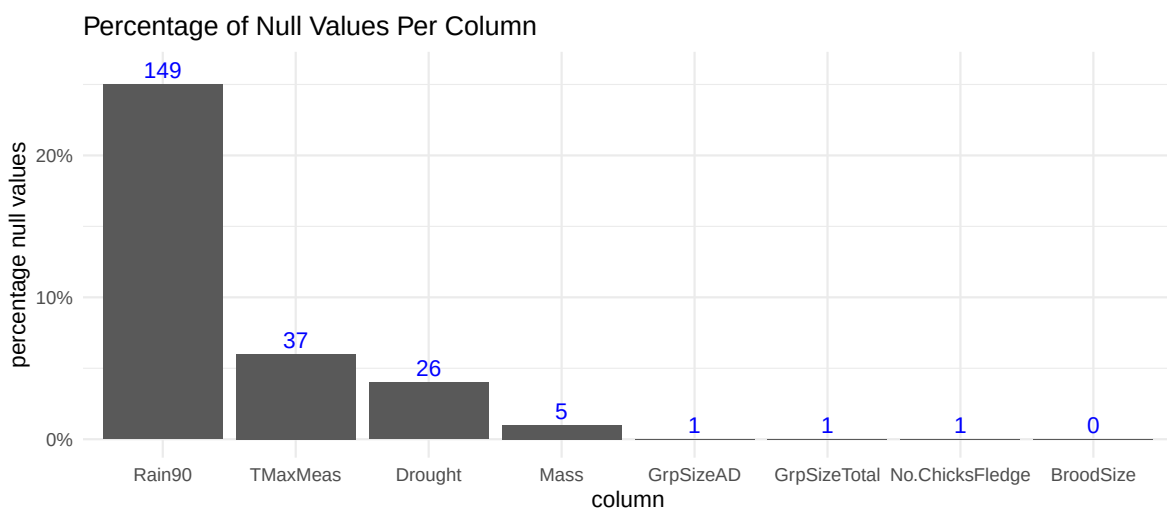
1) ***piedbabbler*** *samplesize per year*

Table 3: Sample Size Per Year

year	sample size
2005	39
2006	60
2007	28
2008	58
2009	33
2010	42
2011	56
2012	45
2013	46
2014	74
2015	30
2016	25
2017	42
2018	18

2)

Null Value Check



7 of 8 selected columns contain at least one null value; the **Rain90** column has the most, with 149 missing values, accounting for ~25% of the data.

Outlier Analysis

Outlier Definition: An outlier is an individual point of data that is distant from other points in the data set. It is an anomaly in the dataset that may be caused by a range of errors in capturing, processing or manipulating data. (**seldon_outliers?**)

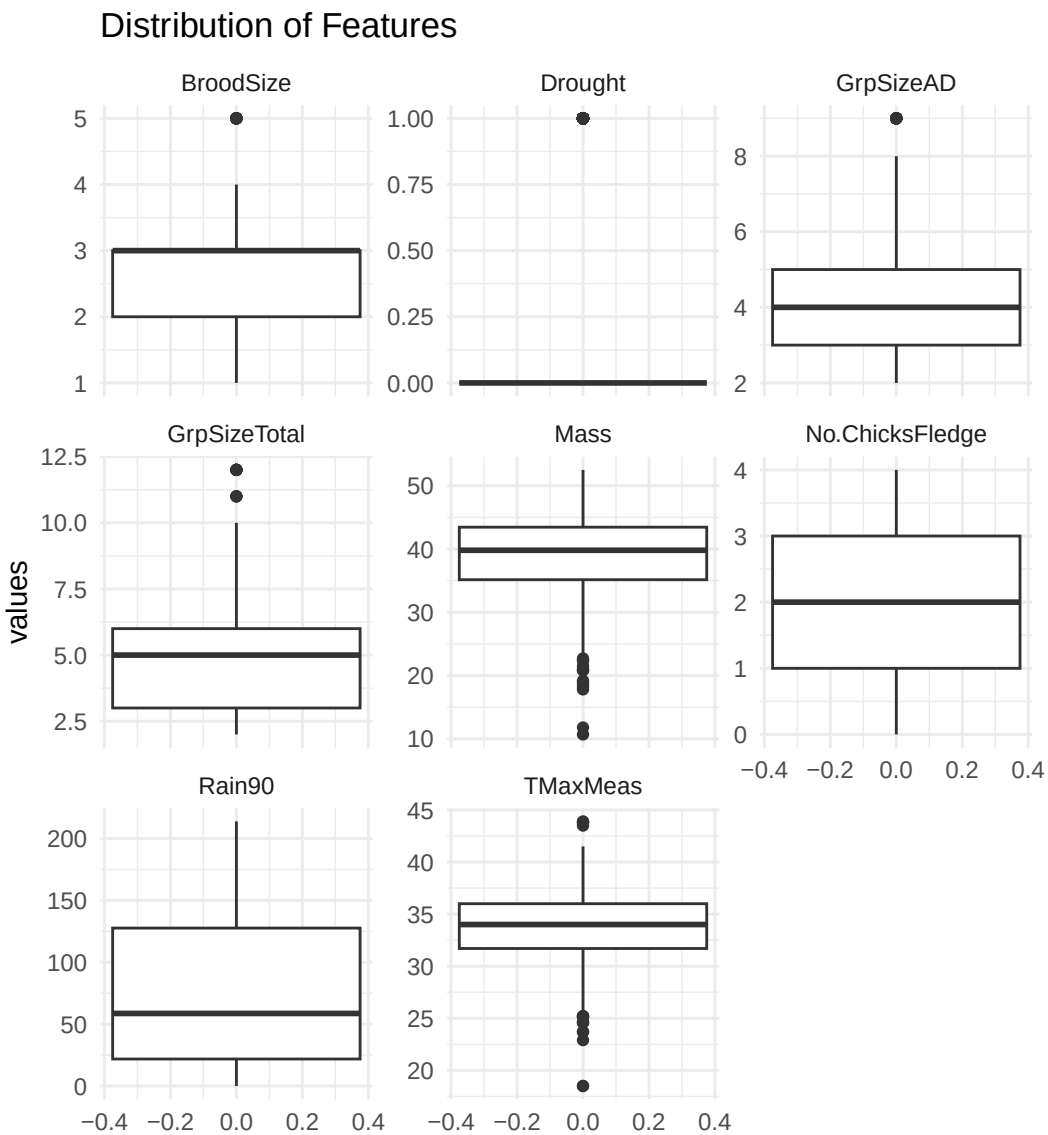


Figure 1: Box Plot Per Feature

Based on their respective boxplots, the following columns appear to contain outliers relative to their individual distributions:

- BroodSize
- Drought
- GrpSizeAD
- GrpSizeTotal
- Mass
- TMaxMeas

3) *Mass and Date per Nest: Longitudinal Dataframe*

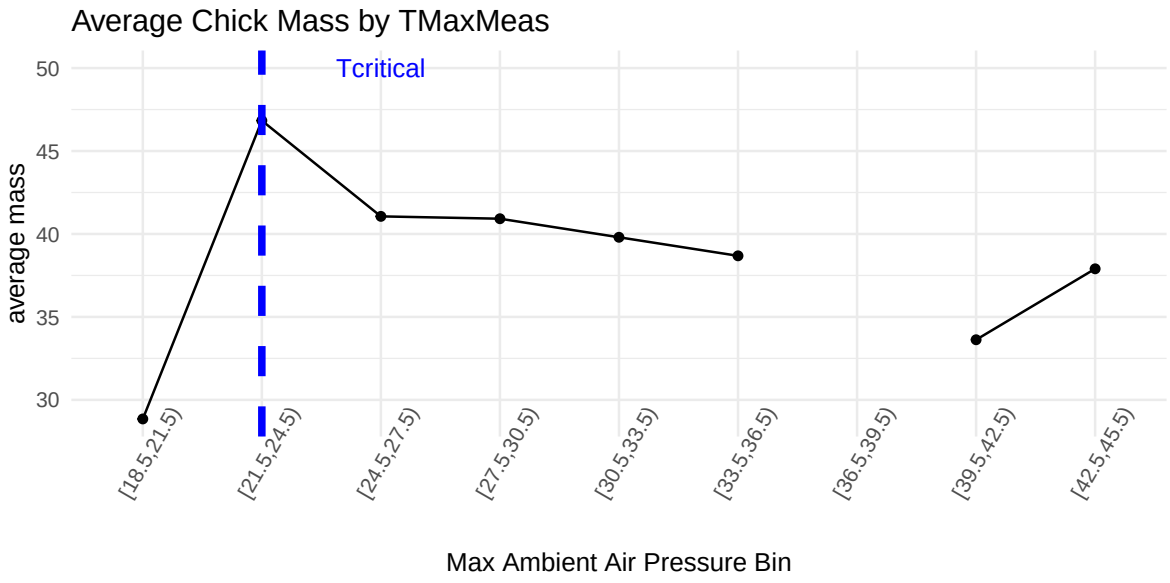
According to the nest_all metadata, the **GRP** column provides the "Group Identity" per bird. The same identifiers are used in the **NestID** column as well. This column will be used as the proxy to identify which nest a bird belongs to going forward. The **sum** is used as the aggregate to calculate mass per nest per **year.**

Nest	2005	2006	2011	2012	2013	2014	2015	2016	2017	2007	2008	2009	2010	2018
AQA	100.1	74.7	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
BAB	63.7	193.2	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
BAL	79.5	68.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
BMD	NA	NA	234.5	140.8	42.8	213.8	108.2	154.0	246.64	NA	NA	NA	NA	NA
CLA	NA	NA	110.4	NA	189.4	82.5	NA	NA	NA	47.1	190.7	108.7	129.2	NA
CMF	NA	NA	NA	NA	121.9	185.7	NA	NA	152.37	NA	NA	NA	NA	NA
CRS	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	88.6	NA	NA	NA
CTR	NA	NA	NA	NA	61.3	140.7	NA	NA	NA	NA	NA	NA	NA	NA
DNV	NA	40.1	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
DTF	NA	NA	110.8	133.1	NA	NA	NA	NA	NA	NA	NA	26.3	121.0	NA
FFL	NA	NA	NA	NA	213.6	NA	NA	NA	NA	NA	NA	NA	NA	NA
GHO	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	112.7
GON	NA	87.3	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
HAR	197.3	153.4	NA	167.1	167.4	208.1	96.2	41.3	162.61	132.4	223.7	221.4	195.6	NA
HLV	NA	NA	NA	70.9	45.5	132.5	NA	NA	NA	NA	NA	NA	NA	NA
HOG	175.1	128.0	236.6	92.9	98.7	NA	NA	NA	NA	90.5	258.8	104.5	136.7	NA
ICE	NA	NA	NA	NA	NA	133.6	65.8	NA	105.57	NA	NA	NA	NA	NA
INF	NA	112.7	129.2	64.9	117.1	295.2	NA	109.2	124.58	153.4	124.0	NA	82.4	NA
JBW	NA	NA	156.7	NA	94.9	163.0	107.7	89.5	143.13	NA	NA	NA	NA	NA
KUR	NA	97.0	NA	NA	NA	NA	NA	NA	NA	80.0	NA	NA	NA	NA
MHC	NA	NA	NA	29.8	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
MLT	NA	NA	NA	NA	NA	195.6	78.0	191.2	44.00	NA	NA	NA	NA	43.6
MMN	NA	NA	NA	120.7	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
MRT	NA	NA	NA	NA	NA	NA	NA	38.9	NA	NA	NA	NA	NA	NA
NOL	NA	NA	NA	NA	NA	NA	23.5	38.6	NA	NA	NA	NA	NA	111.5

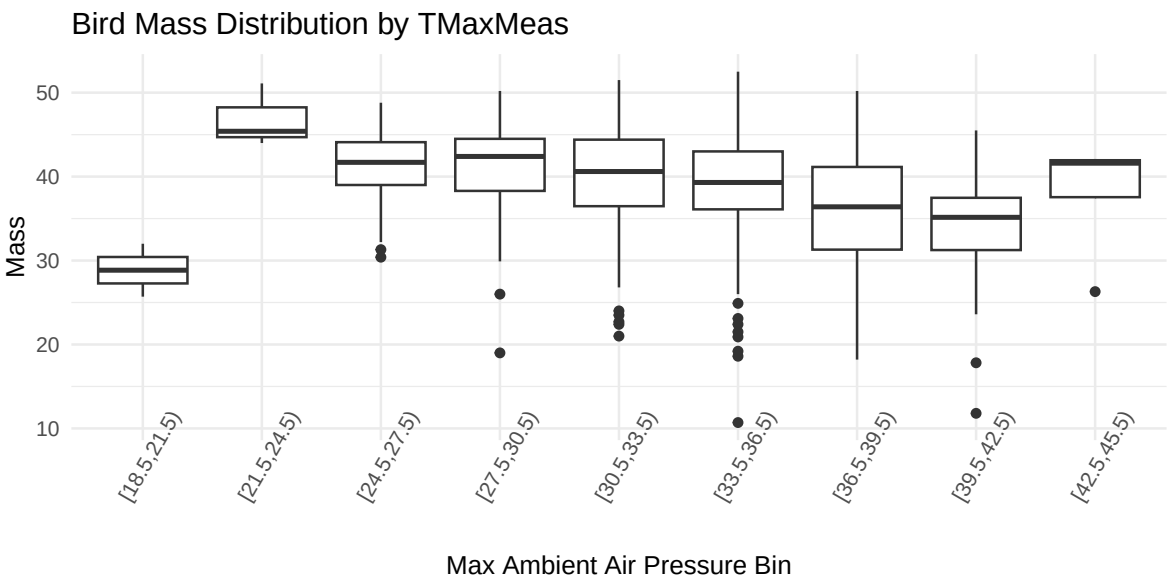
Nest	2005	2006	2011	2012	2013	2014	2015	2016	2017	2007	2008	2009	2010	2018
NTU	NA	NA	201.8	NA	NA	NA	NA	NA	NA	NA	NA	NA	207.8	NA
OLM	NA	NA	NA	NA	119.2	188.7	142.5	NA	198.88	NA	NA	NA	NA	NA
OSD	203.5	345.2	209.6	40.1	120.3	207.8	98.7	NA	NA	196.8	300.2	213.2	197.7	NA
PIX	NA	NA	NA	NA	NA	68.3	NA	NA	NA	NA	NA	NA	NA	71.2
RNB	178.0	206.9	211.5	88.3	159.5	209.0	79.0	85.4	216.72	78.2	337.6	NA	42.8	56.7
RSC	NA	NA	NA	79.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
SEL	NA	NA	NA	NA	NA	NA	NA	NA	47.70	NA	NA	NA	NA	NA
SHA	NA	103.9	88.4	118.5	NA	NA	NA	NA	NA	79.2	96.6	128.6	150.1	NA
SND	NA	NA	168.7	129.9	119.4	117.4	87.9	NA	111.65	NA	115.9	NA	219.0	19.2
SOX	193.7	235.3	51.5	NA	NA	NA	NA	NA	139.56	133.9	436.2	145.2	164.4	NA
SPH	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	46.7	NA	NA
SRC	NA	NA	133.2	139.7	97.5	176.3	66.5	NA	NA	NA	NA	110.1	42.4	NA
STT	NA	NA	NA	NA	34.0	NA	NA	NA	NA	NA	NA	NA	NA	NA
TMT	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	45.2
TNL	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	78.0
TPT	129.6	364.1	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
TRA	NA	NA	NA	76.6	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
TWH	NA	NA	NA	121.0	96.5	NA	NA	NA	NA	NA	NA	NA	NA	NA
UNK	NA	NA	NA	NA	NA	NA	NA	222.6	NA	NA	NA	NA	NA	NA
XHO	141.9	40.4	119.0	166.8	NA	NA	NA	NA	NA	109.5	212.3	145.3	NA	NA

4) *Mass vs Teamparature Tipping Point * might have to check or verify if these are missing at random*

TMaxMeas has 6% of its values as nulls. As such, the aggregate mean Mass value per **binned** values of TMaxMeas will be use to adjust for this. Null values will be replaced with the mean TMaxMeas values



Observing the plot above, the average mass shows a decline when the maximum ambient air temperature exceeds 21.5 degrees (using the minimum value of the bin). The plot below filters specifically for these temperatures.



5. *Mass over time per nest per season*

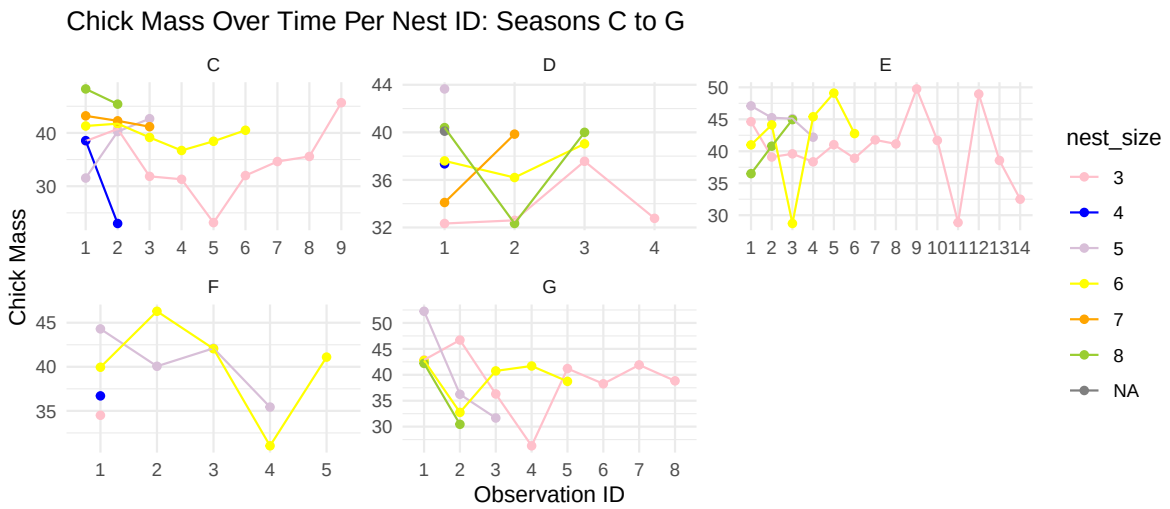
Again, the GRP column will be used as the proxy for the Nest that birds belong to. Since there are over 40 distinct GRP/Nest identifiers, an alternative procedure is used to first filter and identify for "high-mass" nests over time:

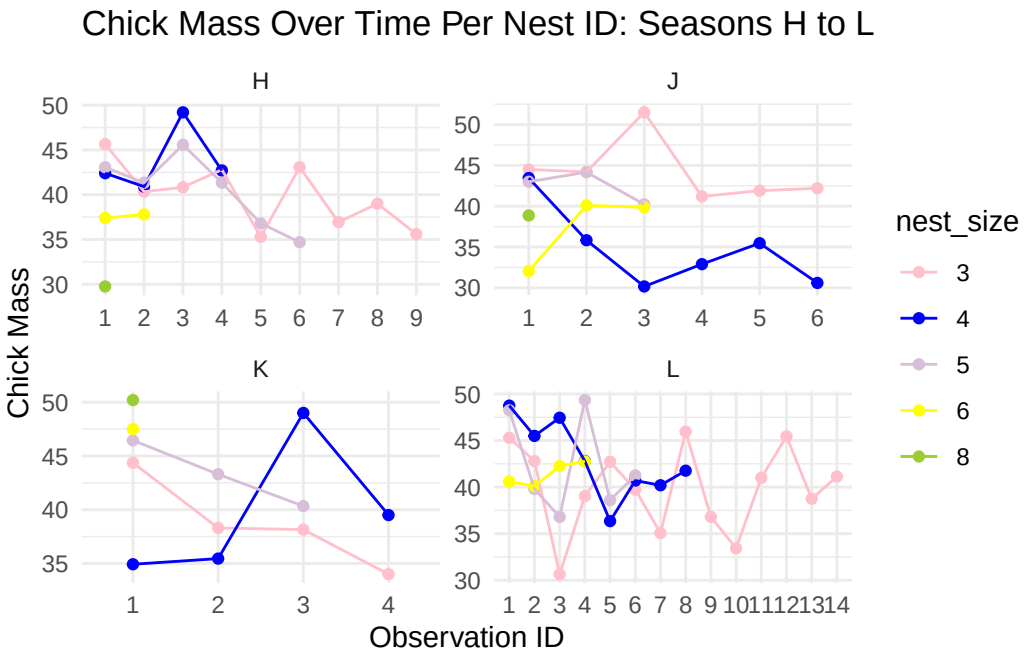
1. First get the size of each nest, defined by the count of birds in each nest, **GrpSizeTotal.** Since this value, changes over time, the **starting nest size** for each group will be used for this.
2. Since the Dates are inconsistent for each group/nest, the Date column is converted to an **Observation ID.** This essentially counts and orderes instances of measurement for each nest, normalizing the time time instances for all groups.
3. Use this Group/Nest as the grouping variable to compare average mass over time over different seasons.
4. Filter for the best per performing nests based on their starting nest size, which hopefully decreases the number of nests/GRP identifiers.
5. Plot the individual average masses over time for these "best performing" nests.

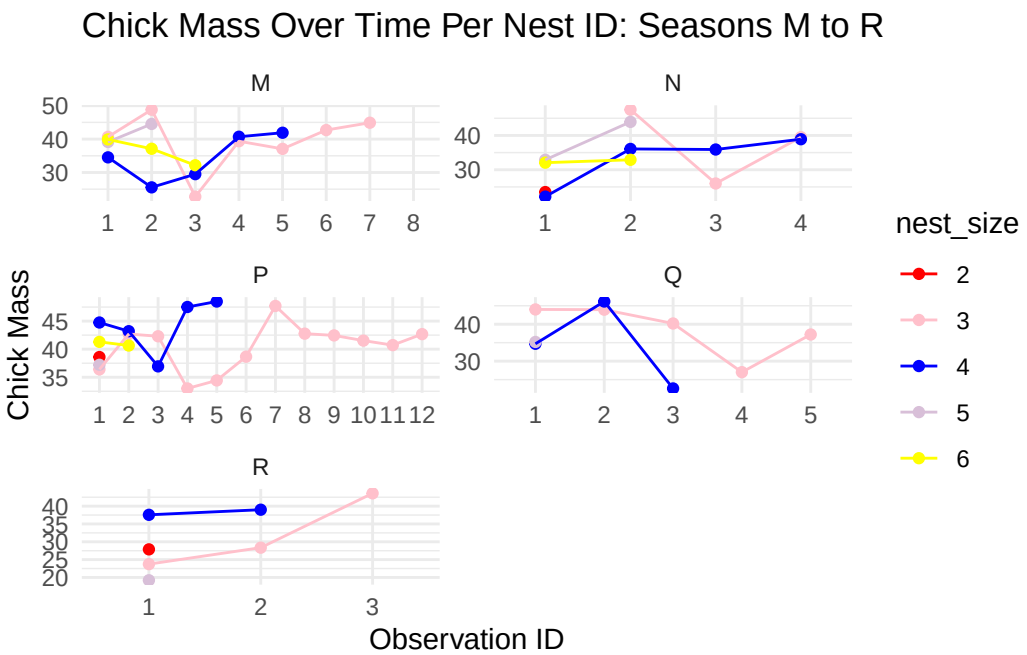
```
# A tibble: 14 x 3
   Season num_index num_index_break
   <chr>      <int> <fct>
 1 C              1 1
 2 D              2 1
 3 E              3 1
 4 F              4 1
 5 G              5 1
 6 H              6 2
 7 J              7 2
 8 K              8 2
 9 L              9 2
10 M             10 3
11 N             11 3
12 P             12 3
13 Q             13 3
14 R             14 3
```

```
# A tibble: 596 x 87
   Season num_index num_index_break Date       BirdID GRP   AgeMeasD Sex
   <chr>      <int> <fct>           <date>     <chr>  <chr>    <int> <chr>
 1 C              1 1               2005-10-03 MLYG   HAR         11 F
 2 C              1 1               2005-10-03 PMPT   HAR         11 F
 3 C              1 1               2005-10-03 TRMB   HAR         11 M
 4 C              1 1               2005-10-04 WRMG   XHO         13 F
 5 C              1 1               2005-10-04 RMPR   XHO         13 M
 6 C              1 1               2005-10-12 PPRM   TPT         11 F
 7 C              1 1               2005-10-12 TRMP   TPT         11 F
```

```
 8 C              1 1                2005-10-12 LMBW   TPT         11 M
 9 C              1 1                2005-10-15 BGPM   OSD         11 F
10 C              1 1                2005-10-15 MBBR   OSD         11 M
# i 586 more rows
# i 79 more variables: NestID <chr>, BirdSeas <chr>, GrpSizeAD <int>,
#   X.Imm <chr>, GrpSizeTotal <int>, TwoMonthsPrior <chr>, IncDate <chr>,
#   HatchDate <chr>, FledgeDate <chr>, FailDate <chr>, IndepDate <chr>,
#   X1YrDate <chr>, X2YrDate <chr>, BroodSize <int>, NoMales <dbl>,
#   BroodSexRatio <dbl>, No.ChicksFledge <dbl>, MaleFledge <dbl>, NoSurv <dbl>,
#   MaleSurv <dbl>, Mass <dbl>, LTars <dbl>, RTars <dbl>, RTarsMm <dbl>, ...
```

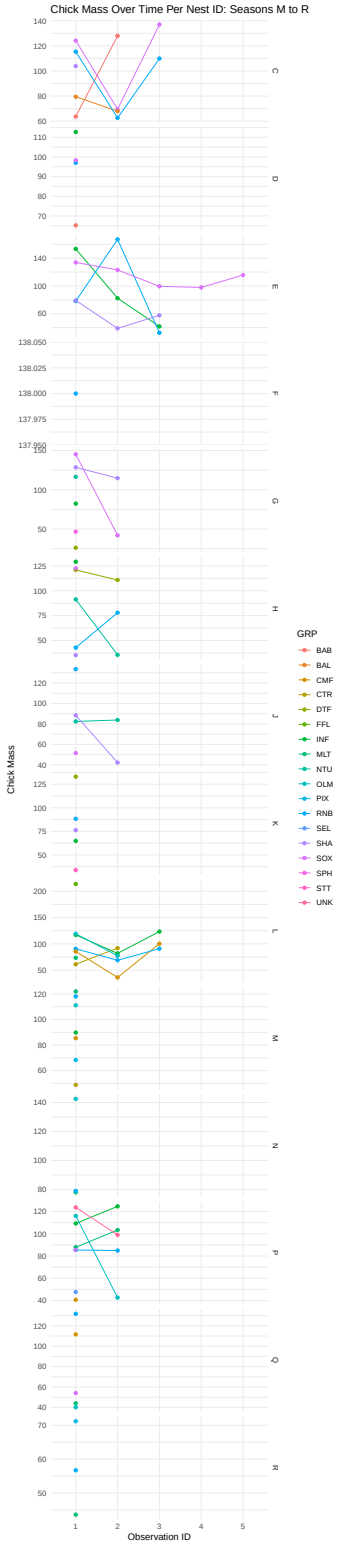




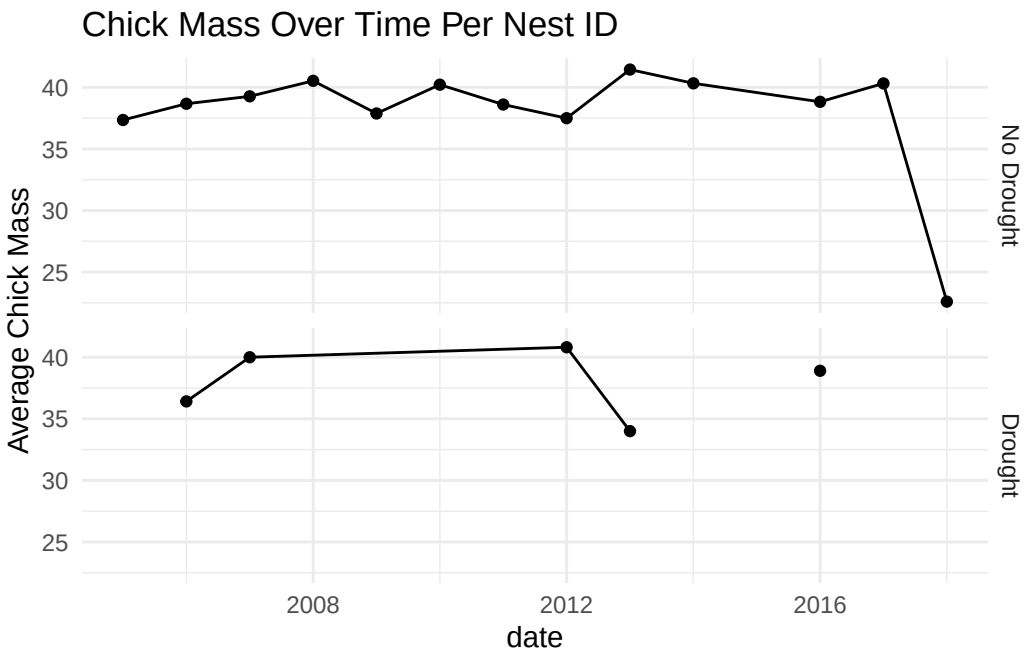


Although the time ranges are inconsistent and disparate across seasons, nest sizes with 3 chicks, show higher masses over time. This will be used as the criterion to filter and observe individual nest ids that fall in this category.

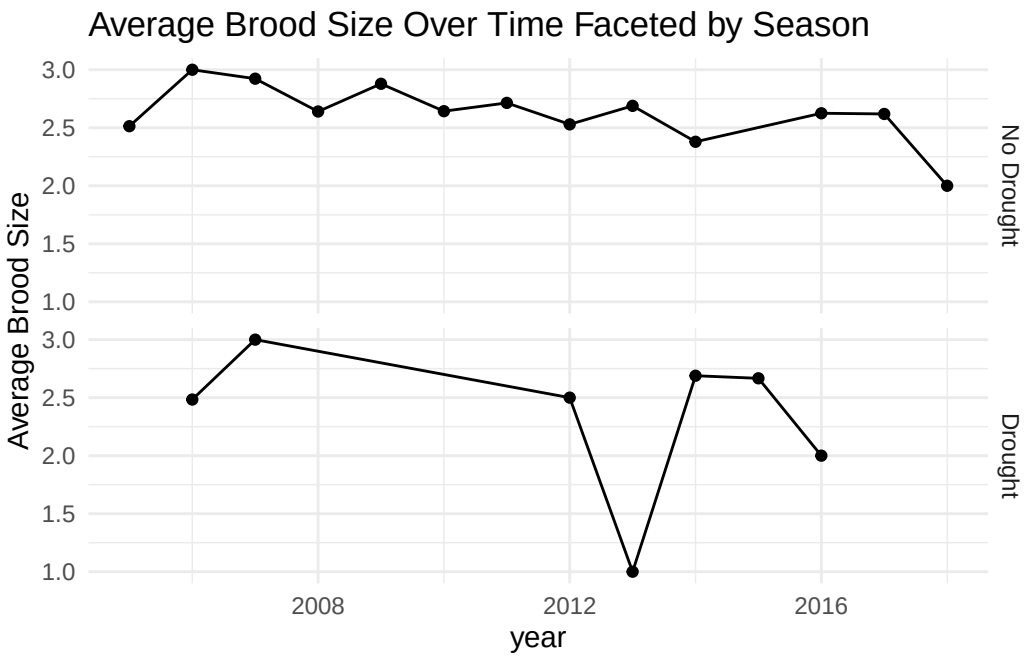
```
# A tibble: 45 x 2
   GRP   nest_size
   <chr>     <int>
 1 AQA           4
 2 BAB           3
 3 BAL           3
 4 BMD           4
 5 CLA           5
 6 CMF           3
 7 CRS           5
 8 CTR           3
 9 DNV          NA
10 DTF           3
# i 35 more rows
```


Chick Mass Over Time Per Nest ID: Seasons M to R
Chick Mass
Observation ID
GRP
BAB
BAL
CMF
CTR
DTF
FFL
INF
MLT
NTU
OLM
PIX
RNB
SEL
SHA
SOX
SPH
STT
UNK

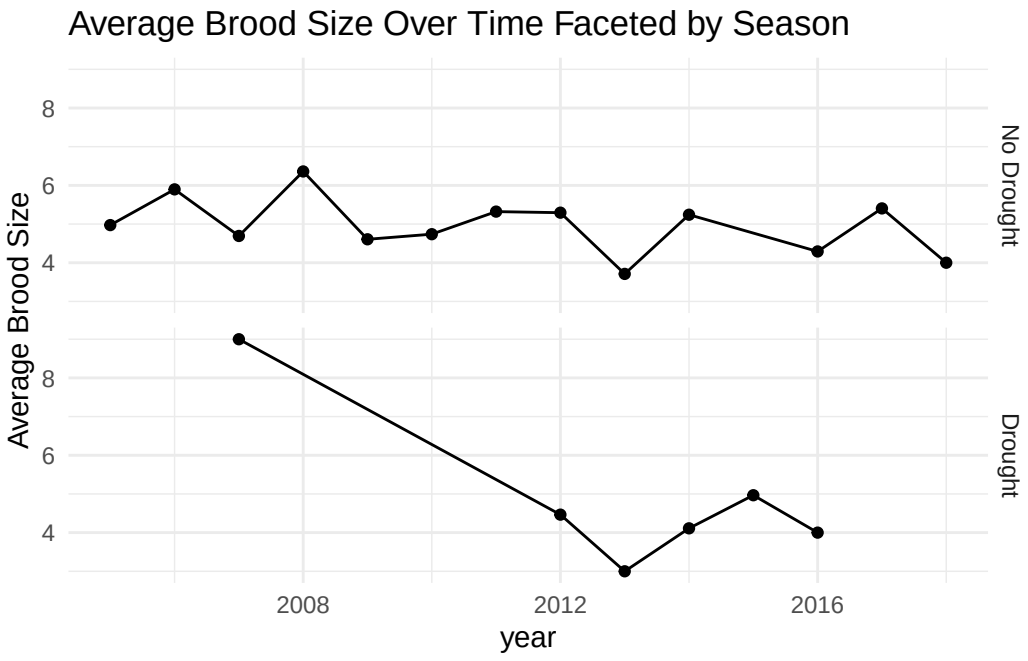
6) *Yearly Average Mass Over Time (Yearly), Faceted By Drought*



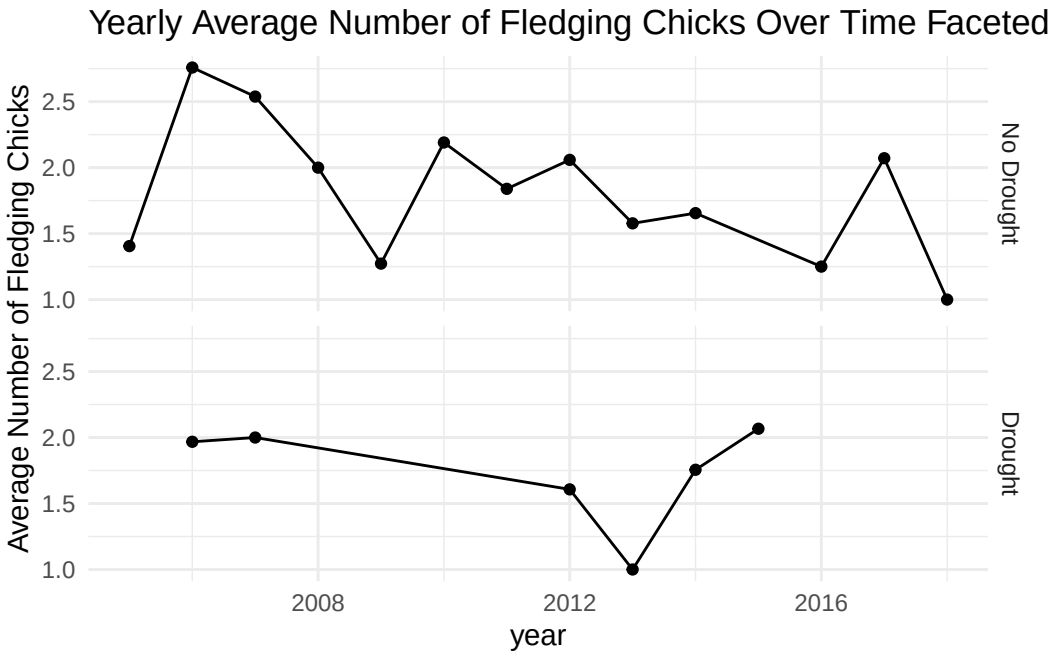
Average Brood Size Over Time (Yearly), Faceted By Drought



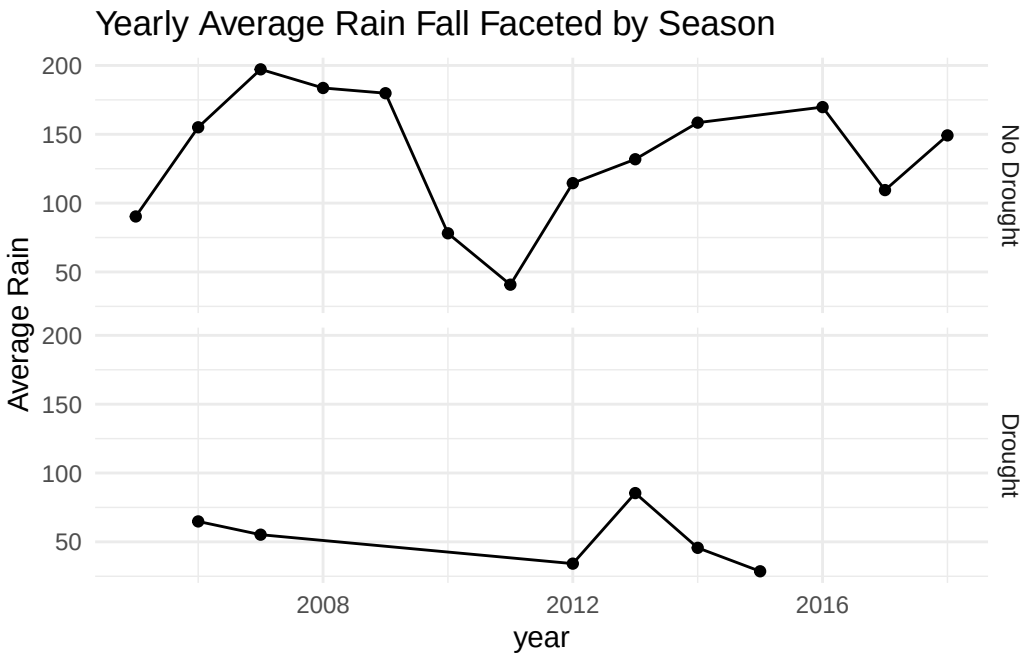
Average Group Size Total Over Time (Yearly), Faceted By Drought



Average Number of Fledging Chicks Over Time (Yearly), Faceted By Drought



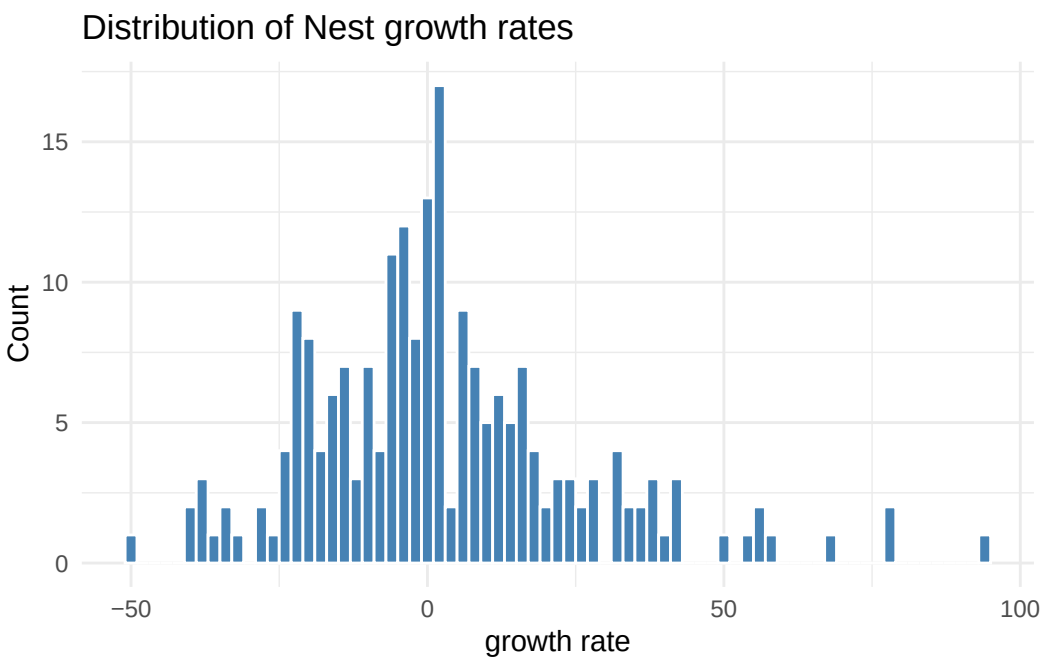
Average Rain Fall Over Time (Yearly), Faceted By Drought



6) *Rate of mass change per day per nesting*

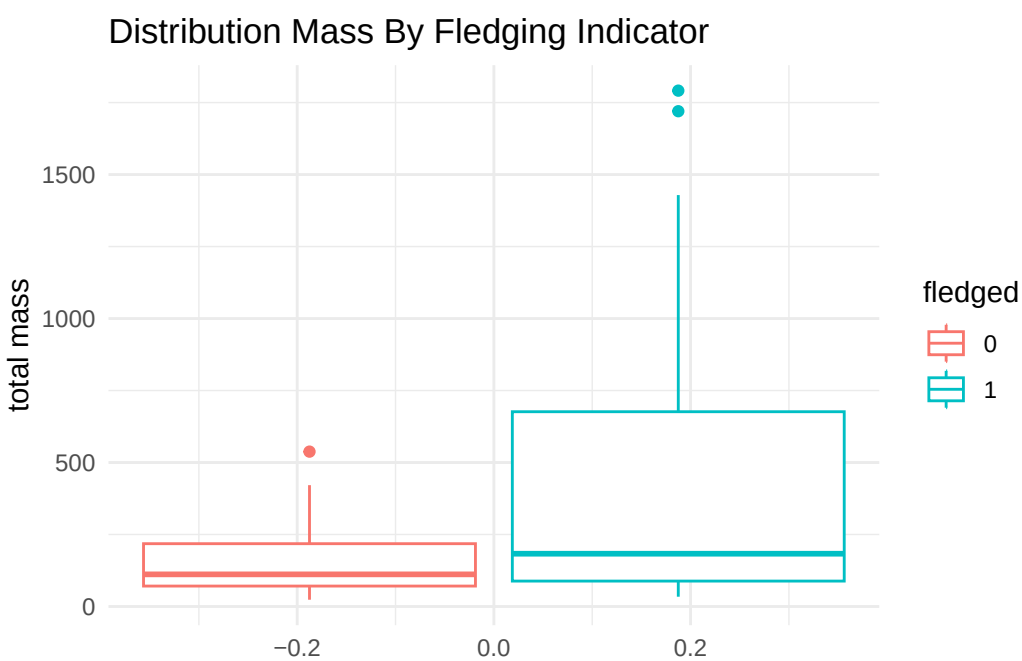
Steps for finding these suspicious entries

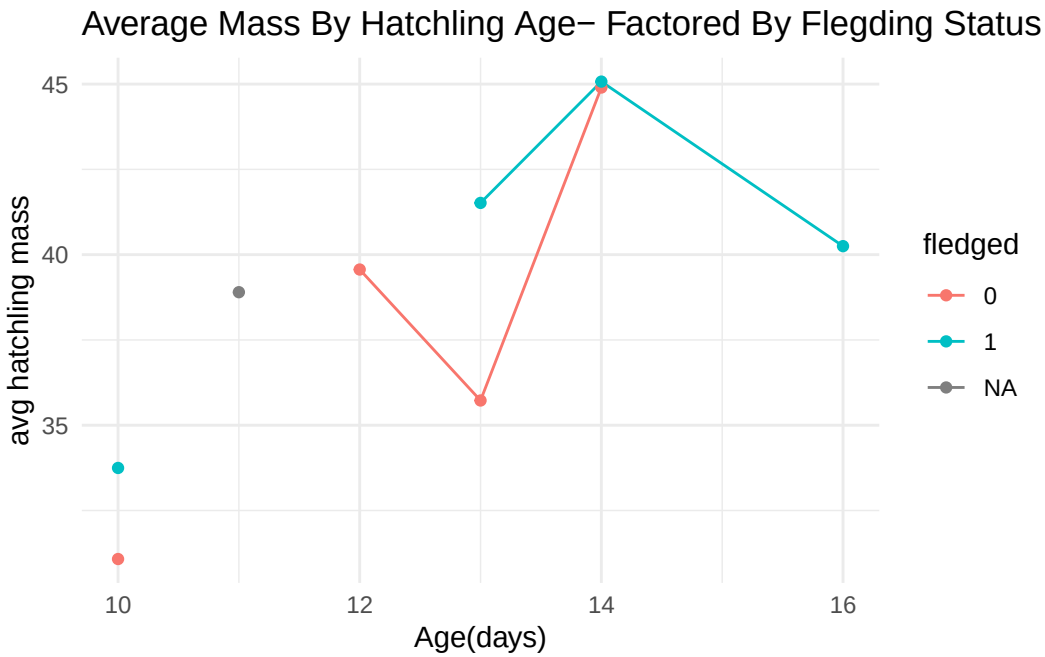
1. First group Mass by GRP and Date to get Mass per Nest over time
2. Get lagged values of Date and Mass
3. Calculate the percentage Mass change for each row per nest group
4. Calculate the date difference, in days, between Date values and lagged date values
5. Filter for date differences equal to or close to 1 then check which growth rates are suspicious (>50%)



Group UNK saw a 25% increase group total mass in just 2 days, although given the distribution of the growth rates across Nests/Groups this does not appear to be anomoloues

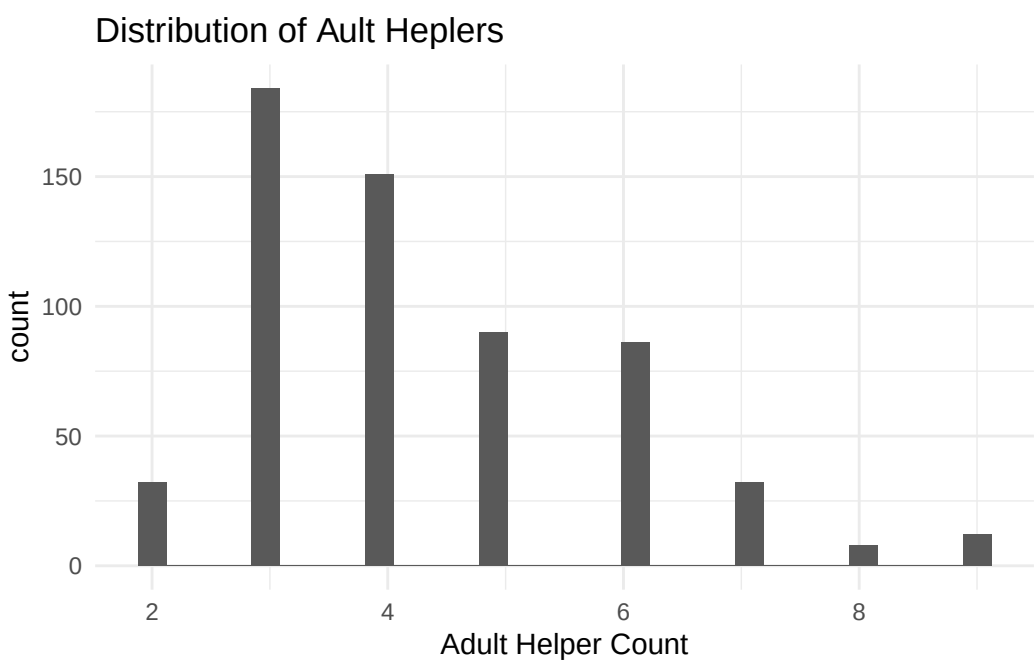
8) *Survivor Bias*





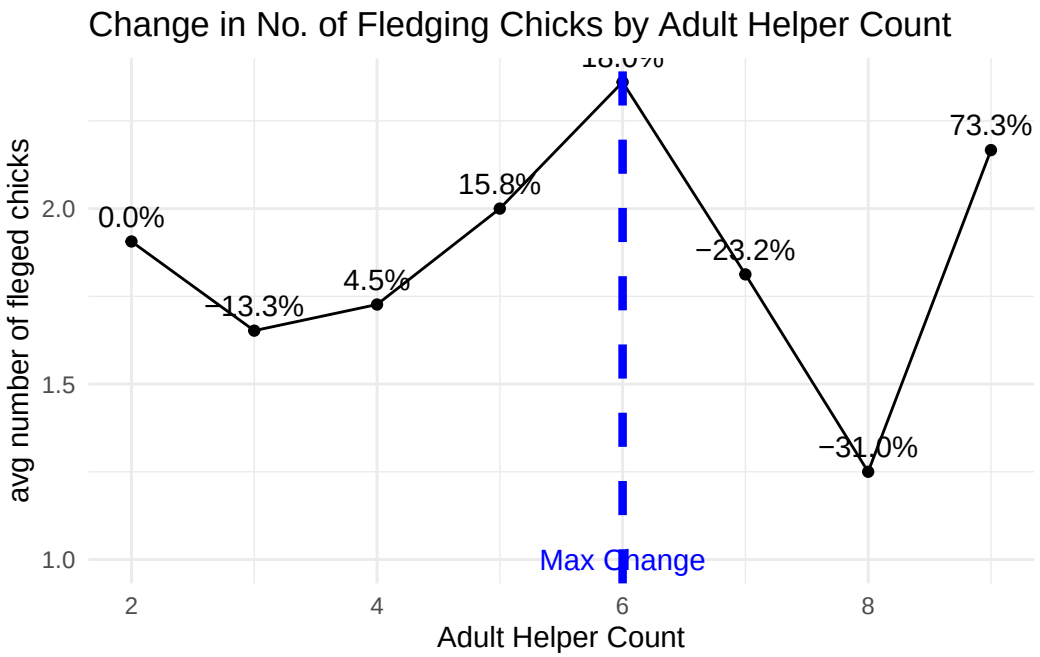
9) *Distribution of Adult Helpers*

First check how the number of adult helpers are distributed across the data



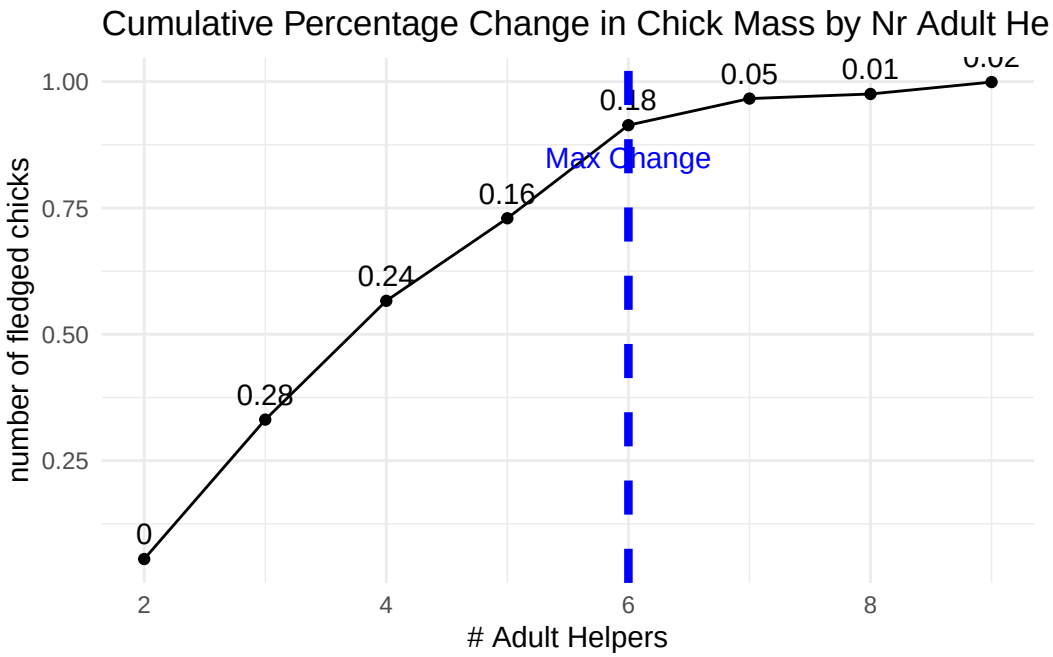
In this case, most nests had adult heplers of 3. Thus, to normalize the data, average fledgling count per adult helper count we'll be use to compare.

9) *Adult Helper Diminishing Return*



The average number fledged chicks sees an increase from 3 to 6, however it sees its most marked sustained decrease from 6 to 8 (although this rebounds after 8)

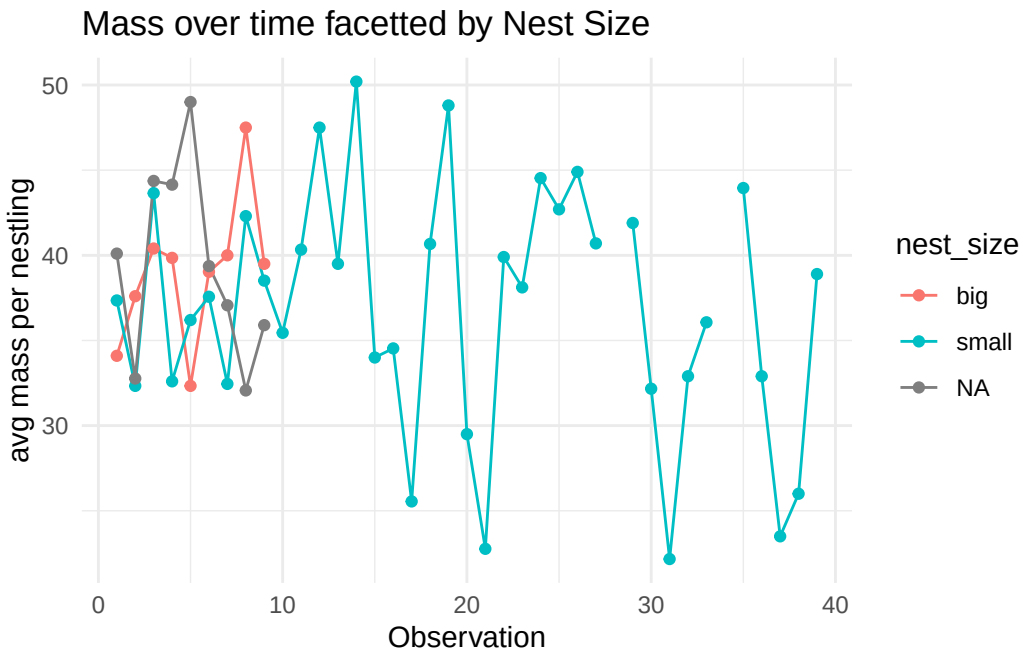
9) *Cumulative View*



10) *Interaction Effect*

Nestiling Mass Per Drought Per Nest Size

- Define big nest size as GrpSizeTotal >6
- Small nest size GrpSizeTotal <6
- Only filtered for periods of drought



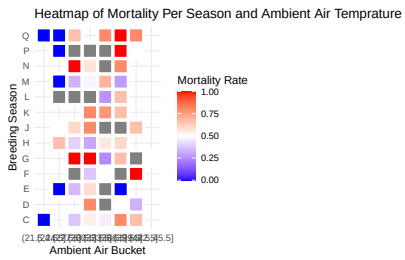
Smaller nests seem to survive longer over periods of drought*

11) *Seasonal Heatmap*

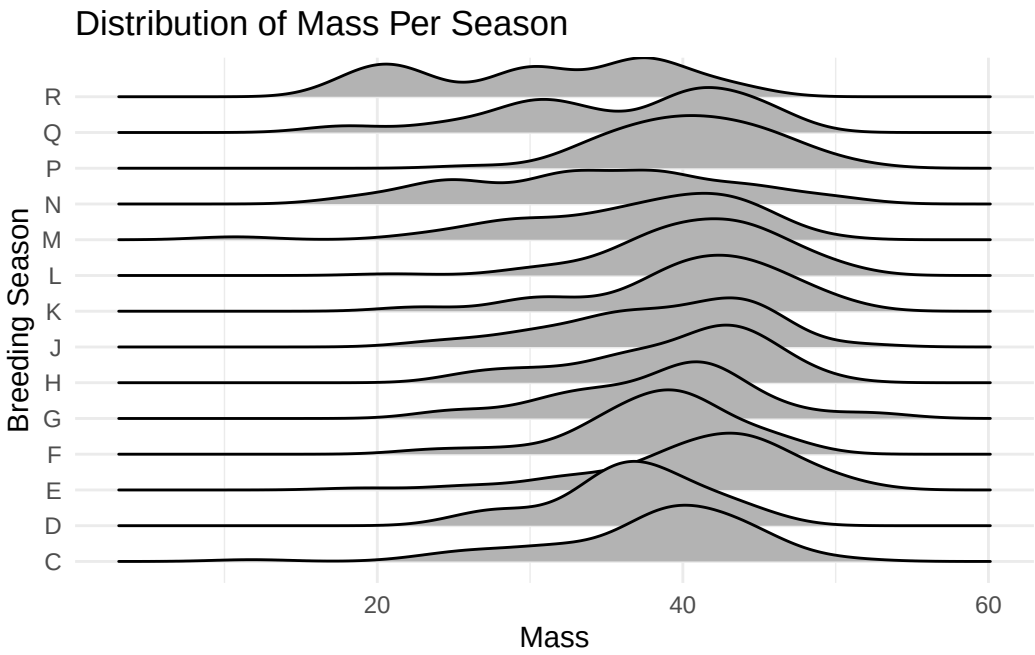
We need to bucket the ambient values

Nesting mortality=number of dead birds/total number of birds

Come back and explain



12) *Ridge Plot*



Season Q has the most bimodal distribution

13) *Survival Flow Sankey Plot*

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[1] 586
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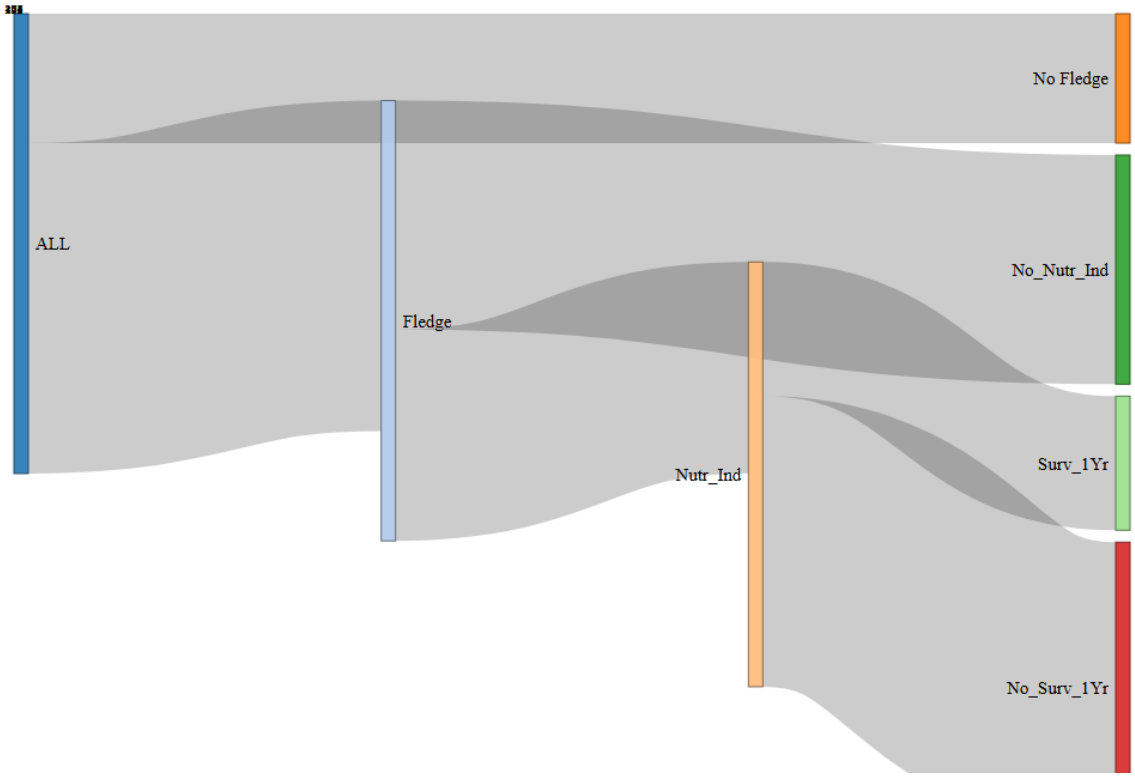
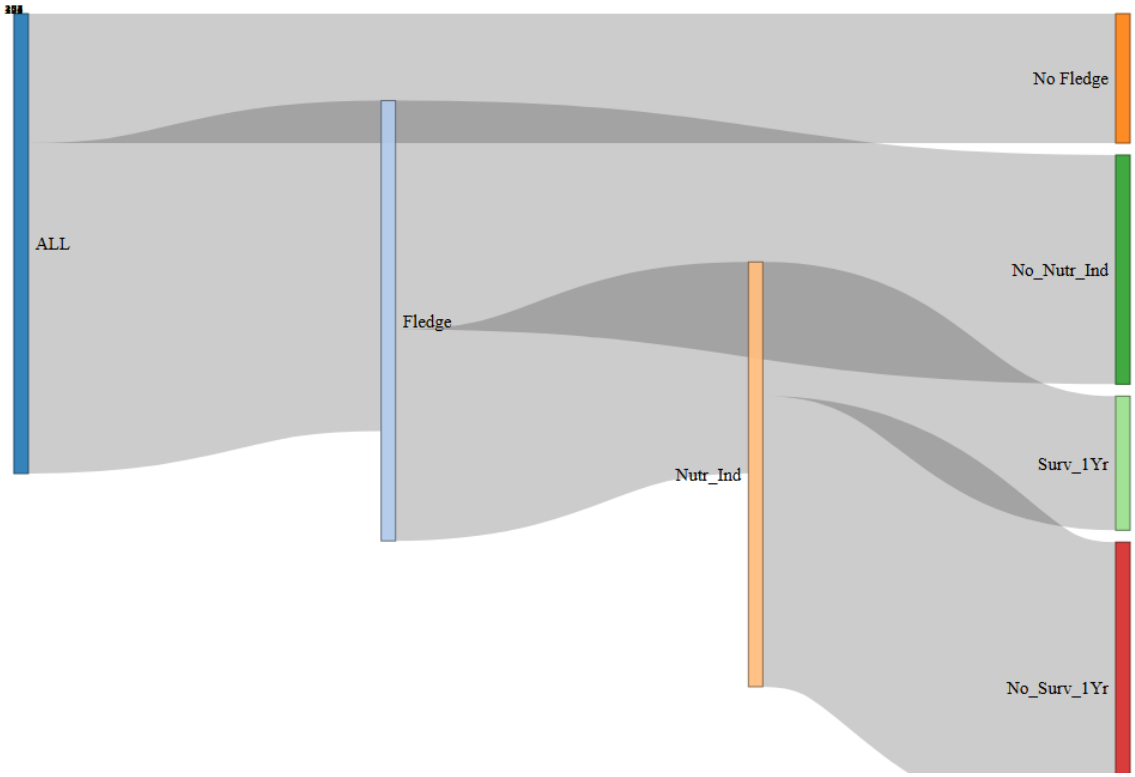
Nestling Survival Flow Sankey Plot



Nestling Survival Flow Sankey Plot